

**Project:**

**Purchase of PTZ camera and accessories for Shek Mun PG Centre  
(11F)**

**Refer no.**

**AV20260202-01-11**

**Submission Deadline:**

**2026-06-26**

**Completion Date:**

**2026-08-02**

## Project requirement:

1. Equipment:
    - a. PTZ Camera with NDI x 12 sets
    - b. PTZ Matrix Tracking Box for 1 channel x 6 sets
    - c. PTZ Matrix Tracking Box for 3 channels x 3 sets
  2. Other:
    - a. HDMI Extender x 19 sets
    - b. Dante adapter x 11 sets
    - c. USB 2.0 extender x 13 sets
    - d. Dante DSP system x 2 sets
    - e. 22U equipment rack x 9 sets
  3. Service:
    - a. Installation service
    - b. Testing report
    - c. Cleaning
  4. Remark:
    - a. The requirement have installation and CAR.
- Reminder:
    - Quotation requirement:
      - 30 days validity
    - Purchase requirement:
      - Include Delivery
      - Provide delivery time
    - Installation or Service requirement:
      - Please confirm to accept or not accept the CAR from HSU. The supplier shall affect their CAR insurance as their own cost if considers not using HSU CAR.(Please remark in quotation.)
    - Payment requirement:
      - 30 days payment after delivery.

# Equipment Specification:

## ■ PTZ camera (with NDI activation) Specification:

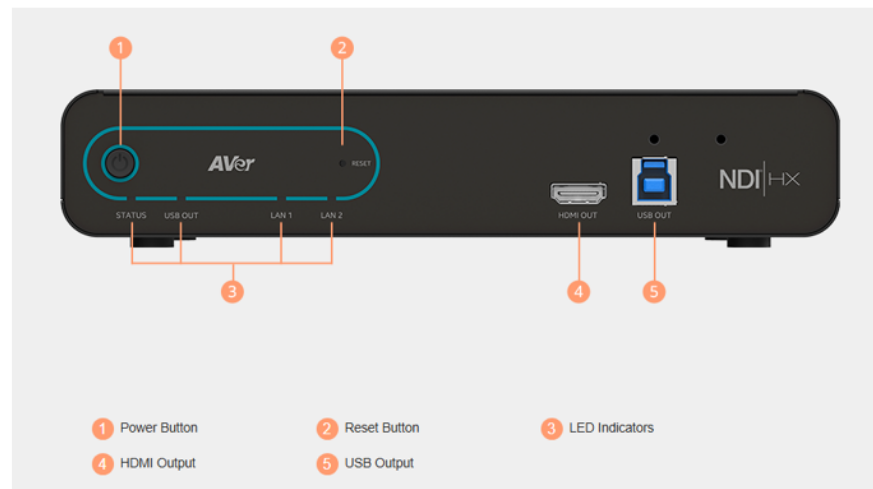
- The PTZ camera (with NDI activation) is a 4K AI-powered PTZR (Pan, Tilt, Zoom, and Rotate) camera designed for professional live streaming. Its built-in NDI|HX3 protocol allows for low-latency, cable-free video transmission over standard IP networks, making it highly versatile for multi-camera broadcasting and remote productions.
  - Core NDI & Networking Specifications
    - NDI Protocol: NDI|HX3 (High Efficiency, low-latency IP streaming).
    - Networking: Gigabit Ethernet (LAN) supporting PoE+ (Power over Ethernet 802.3at).
    - Wireless: Wi-Fi 6 (802.11ax) and Bluetooth 5.4.
    - Compatibility: Seamless integration with standard NDI software and hardware (e.g., OBS, vMix, TriCaster) over your local network.
      - Note: The NDI protocol may require an additional NDI License Key purchase depending on the hardware bundle you select.
  - Key Camera Specifications
    - **Sensor & Resolution:** 1/1.5-inch CMOS sensor, 50 Megapixels. Supports up to 4K@60fps and 1080p@120fps.
    - **Lens:** 22–110mm equivalent, featuring **5x Optical Zoom** and **12x Hybrid Zoom**.
    - **PTZR Mechanics:** 3-axis motorized gimbal with a unique design capable of automated **90° rotation** for instant vertical and horizontal shooting.
    - **AI Features:** AI Tracking 2.0 with upgraded human, animal (over 30 species), and object tracking.
    - **Video I/O & Interfaces:** Includes full-size HDMI, 3G-SDI, USB-C (for UVC streaming), and Ethernet ports.
    - **Battery:** 5000mAh internal battery provides approximately 5 hours of continuous runtime; also supports direct DC power.
    - Included the Tail Air Smart Remote Controller
  - Include remote control
  - Support Wall and ceiling mount.
  - 1-Year Manufacturer Warranty
  - Equivalent or Compatible with OBSBOT Tail 2 (with NDI activation)
- Installation requirement:



# Equipment Specification:

## ■ PTZ Matrix Tracking Box for 1 channel Specification:

- The PTZ Matrix Tracking Box for 1 channel is a powerful, multi-camera AV integration processor designed for hybrid classrooms and business meetings. It supports NDI® HX2 streaming, up to 4K × 60 fps resolution, and is backed by a standard 5-year warranty.
  - Technical Specifications
    - Video Output Resolution: Up to 4K at 60 fps
    - NDI Support: Supports NDI® HX2 (Camera Output) and NDI® HX3 (Camera Input)
    - Connectivity (Inputs): RJ45 LAN 1 (PoE+), RJ45 LAN 2, 2x USB 2.0 (Type-A), 1x HDMI
    - Connectivity (Outputs): 1x USB 3.0 (Type-B), 1x HDMI
    - Display Layouts: Up to 6 different layouts in manual mode
    - Power: PoE+ (Power over Ethernet) or AC 100 - 240V to DC 12V/1.5A
  - 3-Year Manufacturer Warranty
  - Equivalent or Compatible with AVer MT100 Matrix Tracking Box
- Installation requirement:



# Equipment Specification:

## ■ PTZ Matrix Tracking Box for 3 channel Specification:

- The PTZ Matrix Tracking Box for 3 channel (and its NDI-enabled counterpart, the MT300N) is a versatile Matrix Tracking Box designed for Pro AV camera control, multi-device integration, and multi-camera streaming. It simplifies presentation setups by unifying camera feeds, microphone arrays, and displays.
- Key Specifications
  - Output Resolution: Up to 4K at 60 fps
  - Video In/Out: HDMI In/Out and USB In/Out
  - Networking & Control: Supports PoE+ (Power over Ethernet) for single-cable setup and RS-422 control.
  - NDI Support: The MT300N supports NDI®|HX3 inputs and NDI®|HX2 outputs.
  - Software Integration: Built-in PTZ Link Premium for voice-tracking and automatic video switching.
  - Tracking Modes: Supports Live, Manual, and Auto (Microphone/Coordinate-based) tracking modes.
  - UI: Controlled via an intuitive, web-based interface
- 3-Year Manufacturer Warranty
- Equivalent or Compatible with AVer MT300 Matrix Tracking Box

- Installation requirement:



### Equipment Specification:

- HDMI extender Specification:

- The HDMI extender is a professional-grade HDBaseT extender that allows you to transmit signals over long distances via a single Cat5e/6/6A cable. It is specifically built for long-range, stable AV system integration in lecture theaters, conference rooms, and auditoriums, featuring a dedicated audio de-embedded port.
- Key Specifications
- Maximum Distance: Up to 150 meters.
- Audio Support: Features built-in audio de-embedding for routing to external sound systems.
- Transmission Technology: HDBaseT.
- Cable Compatibility: Cat5e, Cat6, or Cat6A
- 1-Year Manufacturer Warranty
- Equivalent or Compatible with VitBest ED-150-TRA

- Installation requirement:



# Equipment Specification:

## ■ Dante Adapter Specification:

- The Dante USB IO Adapter 2x2 allows any PC, Mac, or mobile device to seamlessly connect to a Dante audio network without requiring custom drivers. It features 2x2 bidirectional channels, letting users easily record or play out audio using standard USB or PoE (Power over Ethernet) power.
- Key Features
  - Plug-and-Play: Acts as a class-compliant USB 2.0 audio device; no extra software or drivers are needed.
  - Channel Capacity: Supports 2 inputs and 2 outputs of uncompressed 24-bit / 48 kHz audio.
  - Power Options: Can be powered via USB or a PoE Ethernet switch.
  - Network Compatibility: Fully compatible with Dante Domain Manager (DDM) and supports AES67 for connecting non-Dante networks.
  - Durable Build: Built for the road with a sonically welded ABS enclosure and stress-tested cable assemblies.
- Common Applications
  - Conference Rooms: Provides an easy, drop-in audio connection for laptops and mobile devices (BYOD) to integrate with the room's overarching audio system.
  - Recording/Streaming: Perfect for sending or receiving stereo audio to a Digital Audio Workstation (DAW) over an IP network.
  - Live Events: Reliable and compact enough for AV professionals to throw in a toolkit for quick audio patching
- 1-Year Manufacturer Warranty
- Equivalent or Compatible with Audinate Dante AVIO USB IO Adapter 2x2

- Installation requirement:



# Equipment Specification:

## ■ USB 2.0 extender Specification:

- The USB 2.0 Extender Cat 5 Extender designed for industrial and commercial use. It extends up to 4 USB peripherals up to 50 meters (150 feet) away from your computer via a standard Cat 5 / 5e / 6 cable.
- Key Specifications
  - USB Compliance: USB 2.0 / 1.1 High/Full/Low Speed with data rates up to 480 Mbps.
  - Extension Distance: Up to 50 meters (164 feet).
  - Ports: 1 x USB Type-B Female (Local), 4 x USB Type-A Female (Remote), 1 x RJ45.
  - Housing: Industrial-grade metal mechanical design with integrated wall-mounting.
  - OS Support: Windows, OS X / macOS, and Linux.
  - Power: Local unit is USB bus-powered; remote unit features a DC power adapter to provide sufficient power to connected peripherals.
- 1-Year Manufacturer Warranty
- Equivalent or Compatible with Aten UCE3250

- Installation requirement:



# Equipment Specification:

## ■ Dante DSP system Specification:

- The Dante DSP system is an advanced audio conferencing processor featuring robust IntelliMix DSP algorithms. It comes with a 1-year limited hardware warranty and delivers clear, echo-free sound for modern hybrid meeting and collaboration spaces.
  - Core Specifications
    - Audio Inputs: 10 Dante inputs, 2 analog block connectors, 1 x 3.5mm TRRS mobile jack, and USB.
    - Audio Processing: 8 channels of microphone processing featuring Acoustic Echo Cancellation (AEC), Noise Reduction (NR), and Automatic Gain Control (AGC). Includes IntelliMix automatic mixing, matrix mixing, parametric EQ (PEQ), and compressor.
    - Power Requirement: Power-over-Ethernet Plus (PoE+ / IEEE 802.3at Type 2 Class 4). No outboard power supply needed.
    - Dimensions: 3.6 cm x 13.4 cm x 13.4 cm (1.4 x 5.3 x 5.3 inches).
- 2-Year Manufacturer Warranty
- Equivalent or Compatible with Shure P300-IMX.

- Installation requirement:



# Equipment Specification/Requirement:

## ■ Equipment 22U rack requirement:

- Dimension H1097 x D531 x W531 mm
- 19" rails w/ unit mark, front & rear pairs
- 22U level
- M6 screw fixing, pack of 100
- Includes 4 units of Leveling Caster
- Included fixed Shelf, 50kg Loading x 3 units
- Included 3U 19" Rackmount Drawer with lock x 1 unit
- Included 1U PDU with 8 power sockets (13A) x 2 units
- Included 1U Rack Fan x 1 unit
- Included 3U Fan x 1 unit
- Equivalent or Compatible with Equipment rack

- Installation requirement:



# Equipment Specification/Requirement:

## ■ Mini PC Slot-in PC Module Specification:

- Processor: i7 C10T16 B2.4Ghz M4.9Ghz TDP 45W Cache 24MB
- RAM: 260-Pin DDR4 SODIMM Dual channel DDR4 3200MHz 32GB
- Graphics: Intel® Iris® Xe Graphics/Intel® UHD Graphics
- Storage: NVMe: 256GB/512GB
- Interface: 4×USB3.0, 1×USB Type-C
- Audio: 1×MIC in ( Pink), 1×Line Out ( Green)
- Ethernet: 1×RJ45 (10/100/1000M)
- HDMI: 1×HDMI 2.0, 1×DP1.4
- WLAN: 1×M.2 2230 ( 802.11a/b/g/n/ac/ax 2T2R +BT5.2), 2×2.4G/5G Dual frequency antenna
- Button: 1×Power button, 1×OS reset
- DC power: 12~19V DC
- Operation System: Windows 10/11 64bit
- Power: Power Supply: 80Pin OPS interface power supply
- Input power: 12V~19V
- Max Power Consumption 80.7W
- Dimension: 180×119×30mm
- Net Weight: 0.60kg
- Support: Interactive television
- 1-Year Manufacturer Warranty
- Equivalent or Compatible with OPS72A-i7 13620H

## • Installation requirement:

